



How do I find a closed form of the characteristic function of Gamma distribution?

Asked 5 years, 10 months ago Modified 5 years, 10 months ago Viewed 5k times

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It's written in wikipedia that $(1 - i\beta t)^{-\alpha}$ is the characteristic function of the Gamma distribution $\text{Gamma}(\alpha, \beta)$. I tried to prove this but I failed. So I googled it for a proof, but **surprisingly there is no written proof on google**. Even in this website, there are some questions asking for it, but there is no answer. How do I prove it?

That is,

Let $\alpha, \beta > 0$ and $t \in \mathbb{R}$.

Then, how do I show that $\int_0^\infty x^{\alpha-1} e^{-x(1-i\beta t)/\beta} dx = \Gamma(\alpha) \beta^\alpha (1 - i\beta t)^{-\alpha}$?

Could someone prove this as an answer or tell me a reference which deals with this.

Thank you in advance!

real-analysis integration complex-analysis probability-theory reference-request

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edited Jun 12, 2020 at 10:38

asked Oct 10, 2018 at 11:41



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Rubertos
12.7k 2 31 79

1 ▲
I have to go through all the steps, but I found an alternative proof (not using a contour integral) here:
▼ statlect.com/probability-distributions/gamma-distribution – Sextus Empiricus Nov 18, 2021 at 22:55

1 Answer

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Edit: As pointed out in a comment, this is not quite right. Will be fixed soon, after class...

I said this:

This is just the change of variable $x' = x(1 - i\beta t)/\beta$ plus the definition of $\Gamma(\alpha)$:

$$\int_0^\infty x^{\alpha-1} e^{-x(1-i\beta t)/\beta} dx = ((1 - i\beta t)/\beta)^{-\alpha} \int_0^\infty x^{\alpha-1} e^{-x} dx = \Gamma(\alpha) \beta^\alpha (1 - i\beta t)^{-\alpha}.$$



And of course that's nonsense, it's not just a "change of variable" because $(1 - i\beta t)/\beta$ is not real. It's a simple application of Cauchy's Theorem; some of you can stop reading at this point.



The corrected version:

Let $V = \mathbb{C} \setminus (-\infty, 0]$ be the slit plane, and define $f \in H(V)$ by

$$f(z) = z^{\alpha-1} e^{-z},$$

where $z^{\alpha-1}$ is defined using the principal-value logarithm:

$$z^{\alpha-1} = e^{(\alpha-1) \log(z)}.$$

Note that if $z \in V$ and $x > 0$ then $\log(xz) = \log(x) + \log(z)$, hence

$$(xz)^{\alpha-1} = x^{\alpha-1} z^{\alpha-1}.$$

Also note that if z has positive real part then $f(rz) \rightarrow 0$ exponentially as $r > 0$ tends to $+\infty$.

Set $\omega = (1 - i\beta t)/\beta$ and define $\gamma_1, \gamma_2 : [0, R] \rightarrow \mathbb{C}$ by

$$\gamma_1(x) = x,$$

$$\gamma_2(x) = \omega x.$$

. Let γ_3 be the straight line from R to ωR .

It follows easily from Cauchy's Theorem that

$$\left(\int_{\gamma_1} + \int_{\gamma_3} - \int_{\gamma_2} \right) f(z) dz = 0.$$

Detail: That's not quite just a special case of CT, since our triangle does not lie in V , passing through the origin as it does. That's easily fixed: Consider a contour consisting of that triangle except with a little detour near the origin, so it lies in V . Take a limit. (**Note** that the condition $\alpha - 1 > -1$ is needed to show that the error tends to 0.)

Since f dies exponentially along rays in the right half-plane it follows that

$$\lim_{R \rightarrow \infty} \int_{\gamma_3} f(z) dz = 0,$$

hence

$$\lim_{R \rightarrow \infty} \int_{\gamma_2} f(z) dz = \lim_{R \rightarrow \infty} \int_{\gamma_1} f(z) dz.$$

But

$$\lim_{R \rightarrow \infty} \int_{\gamma_1} f(z) dz = \int_0^\infty x^{\alpha-1} e^{-x} dx = \Gamma(\alpha),$$

while, recalling that $(\gamma_2(x))^{\alpha-1} = \omega^{\alpha-1} x^{\alpha-1}$, the definition of $\int_{\gamma} f(z) dz$ shows that

$$\lim_{R \rightarrow \infty} \int_{\gamma_2} f(z) dz = \omega^{\alpha} \int_0^{\infty} x^{\alpha-1} e^{-x\omega} dx.$$

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edited Oct 11, 2018 at 14:21

answered Oct 10, 2018 at 12:18



David C. Ullrich

91.1k 5 78 145

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- 2 The change of variables (with complex arguments) need some additional verification: You shift the integration region in the complex plane. This is a typical argument, where Cauchy's integral theorem is applied. (Restrict to finite bounds $[a, b]$. Take a box and show that the vertical paths vanish for $a \downarrow 0$ and $b \rightarrow \infty$.) – [p4sch](#) Oct 10, 2018 at 13:53
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- @p4sch You're absolutely right, thanks. I didn't notice the i in the change of variables... – [David C. Ullrich](#) Oct 10, 2018 at 13:58
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- Gamma distribution can be obtained by taking finite sum of iid exponential distributions and characteristic function of the exponential distribution is easy to compute. – [Anil Bagchi](#). Mar 30 at 15:11
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